

AI-based Damage Detection for Power Grid Equipment with SaaS-oriented Design

Midterm Defense

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Project Overview: A Scalable SaaS Inspection Platform

AI Platform Core

- SaaS backend
 - unified auth / RBAC
 - multi-tenant isolation
 - task queue / data lake
- AI model registry
 - Module A: Bird Nest Detection (Done)
 - Module B: Oil Leak Detection (In progress)
 - Module C: Infrared High-Temp. (Integrated)

Multi-modal Inputs

- UAV RGB imagery
- Infrared imagery
- Sensor telemetry streams
- Historical inspection records

Enterprise Outputs

- Alert Center (Dashboards)
- Work Order Dispatch
- Visual Analytics Portal
- API Integration

Midterm Progress Summary

Strategy: “Vertical Slice First, then Horizontal Scale”

Complete one module end-to-end before extending to multi-module registry and unified scheduling.

Module A (Completed)

- Mask R-CNN (Segm.)
- Bbox $AP_{50} = 1.0$
- Flask Deployed
- **Status:** End-to-end PoC

Module B (In Progress)

- YOLOv8s (Det.)
- Trained: CSDN Data
- Best mAP50 ≈ 0.39
- **Status:** API Integrated

Module C (Integrated)

- OCR + Pixel Mapping
- Training-Free
- Custom Alarm Temp.
- **Status:** Visualized & Deployed

Dataset & Augmentation

- **54 COCO Images:** Bbox + Polygons
- **Deterministic Augmentation:**
 - Random IoU Crop (0.3–1.0)
 - Photometric jitter
 - ResizeMax + PadSquare (512×512)
- → *Avoids generative artifacts.*

Model & Config

- **Arch:** Mask R-CNN (ResNet50 + FPNv2)
- **Opt:** AdamW (5×10^{-4}) + OneCycleLR
- **Tech:** AMP Mixed Precision
- **Perf:** $\sim 27\text{s/epoch}$, 70–85% GPU

Module A: Bird Nest Detection — Validation Metrics

COCO Validation Metrics

Metric	bbox	segm
AP@[0.50:0.95]	0.805	0.795
AP50	1.000	1.000
AP75	1.000	0.919
AR@[0.50:0.95]	0.837	0.828

Score threshold 0.4–0.5 gives optimal F1 trade-off in deployment.

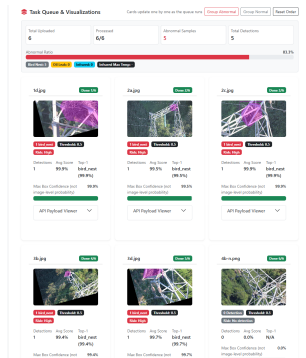


Figure: Bird nest detection: bounding box and instance mask overlay.

Module A: Bird Nest Detection — Training Dynamics

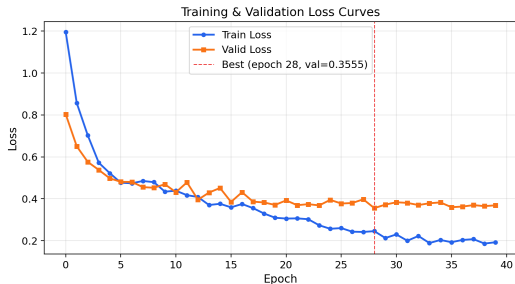


Figure: Train/Validation Loss Curves

Key Observations

- Train: 1.41 $\rightarrow$ 0.14 | Val: 0.65 $\rightarrow$ 0.19
- Plateaus $\sim$ epoch 20 — No Overfitting
- Right-skewed IoU: Strong Localization
- AR/AP drops gracefully at IoU 0.90

Module A: Bird Nest Detection — PR & Threshold Analysis

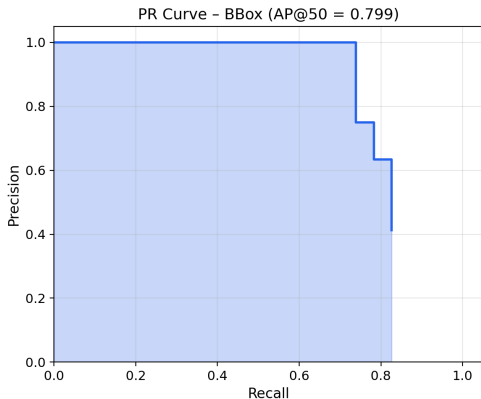


Figure: Bbox PR Curves (IoU=0.50/0.75/0.90)

With $AP_{50}=1.0$, threshold 0.4–0.5 optimizes F1; higher thresholds (0.6–0.8) trade recall for precision in deployment.

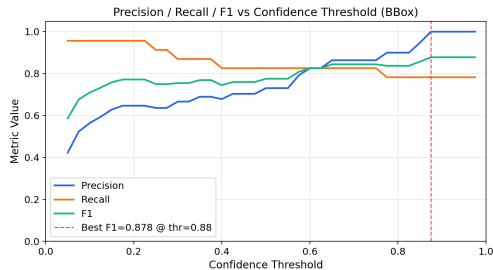


Figure: Score Threshold vs. P/R/F1 (IoU $\geq$ 0.5)

Data Sources

- VOC (~103) + CSDN (~230)
- `prepare_dataset.py` → Unified YOLO format
- Single class: `oil_leak`
- **Current Run (train10):** CSDN Only

Training Config (YOLOv8s)

- **Arch:** `yolov8s.pt` (Single Class)
- **Hyper:** 150 epochs, batch 8, 640×640
- **Opt:** AdamW + Cosine LR
- **Aug:** Mosaic (0.3), Mixup (0.05)

Module B: Oil Leak Detection — Validation Metrics

Best Validation Results (train10-csdsn, epoch 93–94)

Metric	Value
Precision(B)	≈ 0.83
Recall(B)	≈ 0.34
mAP50(B)	$\approx \mathbf{0.39}$
mAP50-95(B)	$\approx \mathbf{0.26}$
Total epochs run	123

→ High precision / low recall indicates conservative detection. Next step: Train on full merged data.

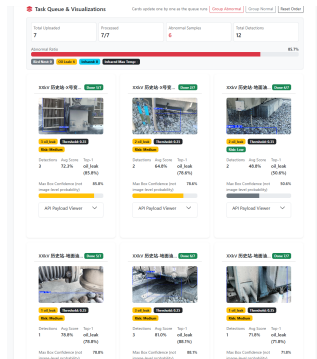


Figure: Oil leak detection: identified leakage region on equipment.

Module B: Oil Leak Detection — Training Results

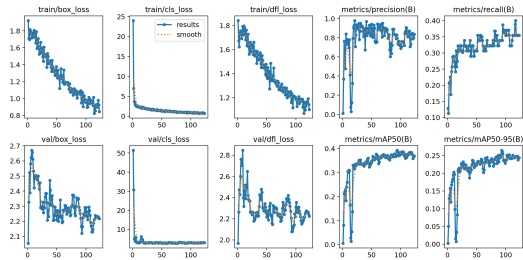


Figure: Training Results (loss / mAP curves)

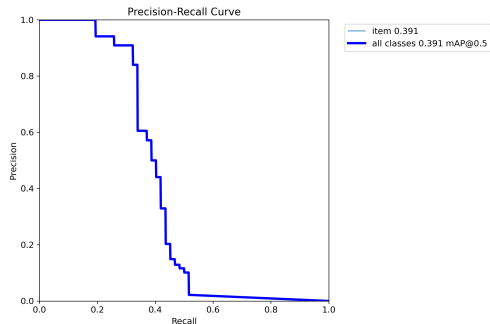


Figure: Validation PR Curve (IoU 0.5)

Module B: Oil Leak Detection — F1 & Confusion Matrix

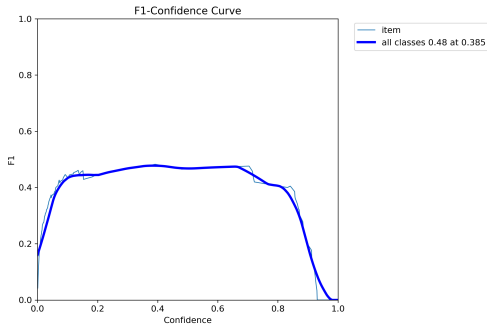


Figure: Validation F1 Curve

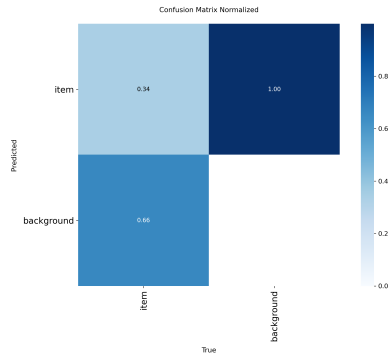


Figure: Normalized Confusion Matrix

→ *Box-level detection identifies location; pixel-segmentation can be added if area tracking is needed.*

Module C: Infrared High-Temperature Detection

Approach: OCR Hybrid

- Deep-Learning Free
- EasyOCR: Parses temp scale
- Pixel-to-Temp Mapping
- **User Parameter:** Custom Alarm (50°C)

Noise Suppression

- **Watermark Mask:** Ignores white logos
- **Legend Mask:** Excludes OCR regions
- **Result:** Outputs bounded hotspots + ΔT

Module C: Infrared Detection — Result

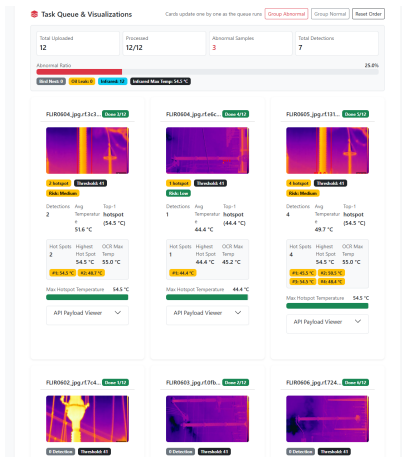


Figure: Hotspot bounding above user-defined threshold.

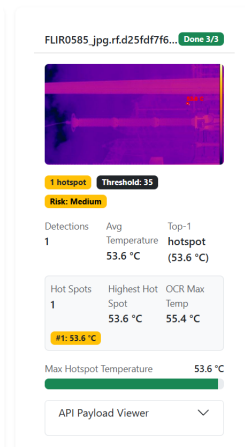
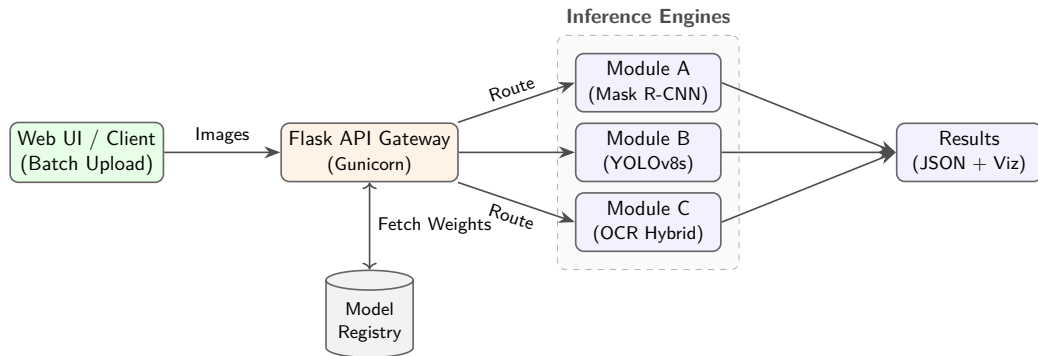


Figure: Infrared detection result card.

SaaS Flask Microservice — Platform Overview

Architecture & Request Flow



SaaS Platform — User Interface

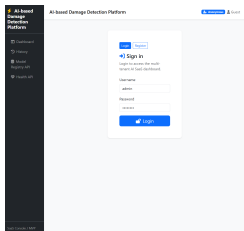


Figure: Login Interface

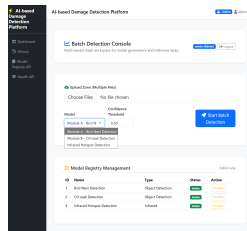


Figure: Main Dashboard & Task Flow

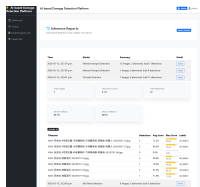


Figure: History & Statistical Analytics for Inference Records and Work Orders

Problems & Challenges

1. Generalization

- **Issue:** $AP_{50}=1.0$ on val set $\rightarrow$ Risk of performance drop in field (haze, new towers).
- **Action:** Integrate cross-scene data & hard-sample mining across the platform.

2. Technical Choices

- **SaaS:** Prioritize multi-model prototype over enterprise auth (OAuth2) for now.
- **Oil Leak:** YOLO box-level suffices; pixel-segmentation only if area stats needed.

3. Constraints

- **Data:** Modest datasets (~ 50 bird nests, ~ 300 oil leaks); annotation is bottleneck.
- **Deploy:** Flask multi-process works; true scale requires queue+worker pool.

Next Phase Work Plan

Track 1: SaaS Backend (Weeks 5–10)

- Multi-tenant RBAC & Task Queue
- AI Model Registry API
- Productize training utilities

Track 2: Modules (Weeks 11–18)

- Oil Leak: Train on merged dataset
- Infrared: Multi-device adaptation
- Reuse data/training pipelines

Integration & Defence (Weeks 19–20)

- End-to-end multi-user testing
- **Final Deliverable:** 3-Module SaaS Prototype + Thesis Report

Discussion Points

- ① **Generalization strategy:** Should the project invest in cross-scene data collection or focus on augmentation and hard-sample mining to improve robustness of all three modules?
- ② **Oil leak improvement:** Current mAP50 ≈ 0.39 on CSDN dataset only. Should training be re-run on the merged VOC+CSDN dataset, or should the model architecture be upgraded (e.g. YOLOv8m / segmentation head)?
- ③ **SaaS depth vs. breadth:** Should the final phase prioritize a fully functional RBAC and task-queue backend, or focus on expanding the AI modules and demo quality for the defence?